

Lab 25 — Radial Basis Function (RBF) Network

Aim

To implement an RBF network with k-means-selected centres for classification on the Iris dataset.

Algorithm / Procedure

1. Load the Iris dataset.
2. Split the dataset into training and testing sets.
3. Apply K-Means clustering to the training data to select RBF centres.
4. Calculate Gaussian RBF activations for each data point.
5. Transform the training and testing data into RBF features.
6. Train a Logistic Regression classifier on the transformed features.
7. Predict the test samples.
8. Calculate and display the classification accuracy.

Program

```
import numpy as np

from sklearn.cluster import KMeans

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score

X,y=load_iris(return_X_y=True)

X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.3,random_state=1)

kmeans=KMeans(n_clusters=10,random_state=1,n_init=10)

kmeans.fit(X_train)

centres=kmeans.cluster_centers_
```

```
sigma=1.0

def rbf_transform(X,centres,sigma):

    return np.array([[np.exp(-np.sum((x-c)**2)/(2*sigma**2)) for c in centres] for x in X])

X_train_rbf=rbf_transform(X_train,centres,sigma)

X_test_rbf=rbf_transform(X_test,centres,sigma)

clf=LogisticRegression(max_iter=1000)

clf.fit(X_train_rbf,y_train)

pred=clf.predict(X_test_rbf)

print("Accuracy:",accuracy_score(y_test,pred))
```

Output

Accuracy: 0.9555555555555556

Result

The Radial Basis Function network was successfully implemented using K-Means clustering to select centres, Gaussian RBF transformation, and Logistic Regression as the output layer.

The model achieved approximately 95.6% accuracy on the Iris dataset.